FOOD COMPOSITION AND ADDITIVES

Determination of Soluble Dietary Fiber in Foods and Food Products: Collaborative Study

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A collaborative study was conducted to determine the soluble dietary fiber (SDF) content of foods and food products by using a combination of enzymatic and gravimetric procedures. The method was basically the same as that employed for determining total dietary fiber (TDF), 985.29, and the method for insoluble dietary fiber (IDF), 991.42. Ten laboratories were each sent 13 test samples (6 blind duplicates and 1 standard [green beans] containing 29–33% TDF, 19–23% IDF, and 8–13% SDF) and were instructed to assay for IDF, SDF, and TDF independently. Included in the package were the 3 enzymes, namely alpha-amylase, amyloglucosidase, and protease, and the filter aid Celite, which was thought to be the major cause of high reproducibility relative standard deviation (RSD_R) values for SDF obtained in a previous collaborative study. The foods to be analyzed were apricots, carrots, chickpeas, onions, raisins, and the sugar beet fiber Fibrex™. IDF, TDF, and SDF were calculated as the weight of residue minus the weight of protein and ash on a dry weight basis. RSD_R values of the IDF results averaged 8.02%, with only 1 food having an RSD_R >10%. The RSD_R values for the TDF results averaged 4.97%, and all foods had an RSD_R <7%. Although the RSD_R values for SDF averaged 14.17%, 4 of the 6 foods had an RSD_R <10%, and 1 of the 2 remaining foods that had a high RSD_R had an SDF content of only 1.2%. In all cases, the RSD_R values of the SDF content of the foods were less than the values for the same foods analyzed in a previous collaborative trial. The enzymatic-gravimetric method for the determination of SDF was adopted first action by AOAC INTERNATIONAL.

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The recommendation was approved by the Committee on Food Nutrition and was adopted by the Official Methods Board of the Association. See "Changes in Official Methods of Analysis" (1994) *J. AOAC Int.* **76**, Jan/Feb issue and "Official Methods Board Actions" (1993) *The Referee*, **17**, July issue.

The enzymatic-gravimetric determination of total dietary fiber (TDF) in foods, **985.29**, was adopted final action by AOAC (1); the enzymatic-gravimetric method for insoluble dietary fiber (IDF), **991.42**, was adopted first action by AOAC (2). In a preliminary study it was found that the basic method for determination of TDF could be modified to measure IDF by filtering out the IDF before precipitating the soluble dietary fiber (SDF) with ethanol; this method was described earlier in a similar method using physiological enzymes (3) and in 2 previous collaborative studies (4, 5).

In the previous study (5), 22 foods were analyzed for SDF. The products had an average SDF value that ranged from 1.35% for chickpeas to 33.42% for prunes. The reproducibility relative standard deviation (RSD_R) values for half the foods analyzed were higher than 20%. Because it was suspected that the large variation was due in part to the variations in the Celite used, extra precautions were taken in the present study to obtain a specific Celite preparation of uniform quality. Also, 5 of the 6 samples chosen for collaborative study were taken from foods that gave high RSD_R values in the previous study (5). The present study was designed according to the rules of Youden and Steiner (6).

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Collaborative Study

The collaborators from the 10 laboratories participating in this study, representing 8 countries, were analysts in food companies, universities, and commercial and government laboratories. Each laboratory was sent 13 test samples (6 blind duplicates and 1 standard) for determination of TDF, IDF, and SDF. The standard, green beans, contained 29–33% TDF, 19–23% IDF, and 8–13% SDF. These values were derived from a previous collaborative study. The collaborators were also sent the 3 enzymes used in the method, namely, alpha-amylase, amyloglucosidase, and protease, as well as the filtering aid Celite. The collaborators were further instructed to weigh test portions to the nearest 0.1 mg and to calculate % TDF, % IDF, and % SDF to 2 decimal places according to the formulas provided.

The following 6 foods were to be analyzed for TDF, IDF, and SDF: (*a*) apricots, (*b*) carrots, (*c*) chickpeas, (*d*) onions, (*e*) raisins, and (*f*) sugar beet fiber. Items (*a*) and (*e*) were supplied by Vacu-Dry, Santa Rosa, CA; item (*b*) was supplied by California Vegetable Concentrate, Modesto, CA; item (*c*) was purchased as the dried material in a local supermarket; item (*d*) was supplied by Basic American Foods, San Francisco, CA; and item (*f*), sugar beet fiber (Fibrex), was graciously donated to us by Delta Fibre Foods, Minneapolis, MN.

To prepare test samples, all products were homogenized in water in a kitchen food processor, lyophilized, ground in a continuous-grinding Microjet 10 Centrifugal Mill (Quartz Technology, Inc., Westbury, NY) to a uniform size, and treated as described previously (5). None of the test samples contained more than 10% fat; therefore, fat extraction was not recommended.

The collaborators performed a moisture analysis before the determination of TDF, IDF, and SDF so that the results could be reported on a dry matter basis (Table 1).

993.19 Soluble Dietary Fiber in Food and Food Products—Enzymatic-Gravimetric Method (Phosphate Buffer)

First Action

[Applicable to determination of soluble dietary fiber (SDF) in vegetables, fruit, and cereal grains and to determination of total dietary fiber (TDF) in conjunction with **991.42**, Insoluble Dietary Fiber (IDF) in Food and Food Products.]

Method Performance

See Tables **993.19A**, **993.19B**, and **993.19C** for method performance data.

(*Caution*: See *Appendix: Laboratory Safety* for safe handling of organic solvents.)

A. Principle

Duplicate test portions of dried foods, fat-extracted if >10% fat, are gelatinized with heat-stable *alpha*-amylase and then enzymatically digested with protease and amyloglucosidase to remove protein and starch. IDF is removed by filtering and washing residue with water. SDF in filtrate is precipitated by adding 95% ethanol to filtrate. Precipitate is filtered and washed with 78% ethanol, 95% ethanol, and acetone; dried; and weighed. One duplicate is analyzed for protein, and the other is incinerated at 525° to determine ash. SDF is weight of residue minus weight of protein and ash.

B. Apparatus

See **991.42B**.

C. Reagents

See **991.42C** with following change: Reagent (**j**), Celite, medium grade (acid-washed) (Sigma Chemical Co., St. Louis, MO, is suitable source).

D. Enzyme Purity

See **991.42D**.

E. Preparation of Samples

Analyze dry foods without pretreatment whenever possible. Dry-mill to 0.3–0.5 mm mesh. Homogenize and freeze-dry wet foods before milling. If high fat content (>10%) prevents proper milling, defat with three 25 mL portions of petroleum ether/g food before milling. Determine residual moisture in milled foods by drying overnight in 70° vacuum oven or 5 h in 105° air oven. Record weight loss due to fat and/or water, and make appropriate correction to final % TDF, IDF and SDF. (*Note*: For foods high in sugars that cannot be dried by lyophilization, extract test sample 3 times each with 10 volumes 85% methanol to remove sugars before milling or lyophilization, which may interfere in determination.)

F. SDF Determination

Proceed as in **991.42F**, from beginning up through "Wash residue...2 times with 10 mL acetone." in paragraph 5.

Adjust weight of combined filtrate and water washings to 100 g with H_2O. Add 4 volumes (400 mL) 95% ethanol, preheated to 60°. Let precipitate form at room temperature 60 min.

Tare crucible containing Celite to nearest 0.1 mg; then wet and redistribute Celite bed in crucible, using stream of 78% EtOH from wash bottle. Apply suction to crucible to draw Celite onto fritted glass as even mat.

Filter precipitate mixture and wash residue successively with three 20 mL portions of 78% EtOH, two 10 mL portions of 95% EtOH, and two 10 mL portions of acetone.

Proceed as in **991.42F**, starting with "Break surface film..." in paragraph 5 through end of **991.42F**.

G. Calculation

Determination of blank:

$$B = \text{blank, mg} = \text{wt residue} - P_B - A_B$$

where wt residue = av. of residue wts (mg) for duplicate blank determinations; and P_B and A_B = wts (mg) of protein and ash, respectively, determined in first and second blank residues.

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Table 1. Collaborative study results (blind duplicates) of determination of total dietary fiber (TDF), insoluble dietary fiber (IDF), and soluble dietary fiber (SDF) by enzymatic-gravimetric method

Lab.	TDF, %		IDF, %		SDF, %	
Apricots						
1	24.68	23.36	—[a]	—	—	—
2	25.61	24.98	12.84	13.82	11.98	12.25
3	26.00	24.26	15.60	13.68	11.23	—
4	25.67	25.88	—	—	—	—
5	25.06	24.82	13.49	14.11	11.74	11.24
6	24.23	24.85	13.31	13.28	11.53	11.57
7	25.19	25.24	13.45	15.20	10.98	9.90
8	24.76	24.56	12.72	12.68	12.40	12.24
9	24.48	22.39	15.45	15.93	10.47	10.56
10	22.79	23.82	13.17	13.56	9.48	10.47
Carrots						
1	22.99	22.43	14.31	13.56	9.01	10.44
2	23.88	23.50	10.31	12.28	12.60	11.28
3	23.24	21.39[b]	14.24	13.26	10.75	11.32
5	24.03	23.89	13.95	14.15	11.29	11.01
6	23.15	23.37	11.52	11.59	12.20	12.20
7	23.46	23.82	12.31	12.60	11.93	11.64
8	24.00	24.12	11.32	11.36	13.56	13.28
9	23.38	23.56	14.13	13.36	10.61	10.76
10	21.92	22.33	11.60	12.03	12.21	11.39
Chickpeas						
1	17.38	14.65	17.45	13.34	0.95	—
2	14.35	13.31	15.35	15.07	1.41	0.65
3	13.78	14.76	16.60	15.75	0.86	1.00
4	14.01	14.57	13.07	14.48	0.81	0.99
5	15.33	14.34	18.93	15.93	1.16	—
6	13.42	14.42	11.80	11.66	1.83	1.70
7	13.40	14.16	15.34	16.49	1.23	1.05
8	13.26	12.88	11.96	11.70	1.64	1.54
9	14.46	14.12	12.32	13.95	1.49	1.40
10	15.65	14.30	16.54	16.59	1.01	1.46
Onions						
1	17.23	15.84	12.42	11.63	4.14	3.63
2	16.38	16.68	12.48	13.50	2.91	3.83
3	16.07	16.16	12.31	11.86	5.21	2.00
5	16.23	17.68	11.73	11.95	5.22	5.35
6	16.15	16.54	13.54	13.14	3.95	3.74
7	16.75	16.22	12.40	13.43	4.25	3.77
8	16.58	16.26	—	—	—	—
9	12.79	16.14[c]	11.22	11.85	5.45	5.26
10	15.56	15.03[c]	12.62	14.66	4.24	3.07
Raisins						
1	31.15	—	27.13	—	7.54	—
2	29.97	28.70	23.22	22.38	—	—
3	31.22	30.73	22.65	22.23	7.46	7.71
4	33.00	34.69	22.26	24.08	7.86	7.38
5	30.22	30.76	21.95	21.40	8.75	8.57
6	29.17	29.87	21.83	22.97	8.56	7.55
7	31.64	31.41	22.05	23.96	9.24	7.23
8	30.20	30.22	22.40	22.22	7.78	7.76
9	29.42	29.60	22.37	21.38	7.64	8.57
10	27.54	25.01	24.56	22.53	4.97	4.02[d]
Sugar beet fiber						
1	67.60	64.30	45.26	46.83	20.25	18.12
2	65.98	65.49	45.66	45.60	22.26	21.22
3	66.39	66.12	44.46	44.28	21.71	21.96
4	68.80	68.27[c]	46.80	47.13	20.44	19.61
5	66.74	66.74	46.78	46.44	21.80	20.39
6	65.80	65.97	44.71	44.92	21.91	22.19
7	65.63	68.03	45.09	46.38	22.49	20.66
8	65.94	65.86	44.96	45.00	19.24	19.94
9	64.50	67.38	46.33	45.53	20.96	20.52
10	63.49	62.49[c]	43.57	45.58	18.93	18.31

[a] No results reported.
[b] Cochran outlier, used in statistical calculations.
[c] Paired Grubbs outlier, used in statistical calculations.
[d] Grubbs outlier, not used in statistical calculations.

Calculate SDF as follows:

$$SDF, \% = \frac{\text{wt residue} - P - A - B}{\text{wt sample}} \times 100$$

where wt residue = av. of wts (mg) for duplicate sample determinations; P and A = wts (mg) of protein and ash, respectively, in first and second sample residues; and wt sample = av. of 2 sample wts (mg) taken.

Calculate TDF as follows:

$$\text{TDF}, \% = \text{SDF} + \text{IDF(from } \mathbf{991.42})$$

Ref.: JAOAC **77**, May/June issue (1994)

Results and Discussion

The determination of TDF by an enzymatic-gravimetric method was adopted final action by AOAC in 1986 and modified in 1988 (4). The modified method was also given final approval by the American Association of Cereal Chemists in 1992 (7) and used in a collaborative study for IDF and SDF in which the IDF method was adopted first action by AOAC (5). In that study of 22 foods, the determination of SDF showed promise; 11 of the foods had RSD_R values <20%. However, the remaining foods had high RSD_R values that could not necessarily be associated with low SDF results. Preliminary results for the SDF determination indicated that a major reason for the differences between laboratory results was the long and vari-

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Table 993.19A. Method performance for soluble dietary fiber (SDF) in foods by enzymatic-gravimetric method (phosphate buffer)

Food	No. of labs	SDF av., %	s_r	s_R	RSD_r, %	RSD_R, %
Apricots	8	11.20	0.42	0.91	3.78	8.11
Carrots	9	11.53	0.53	1.11	4.59	9.61
Chickpeas	10	1.21	0.23	0.34	19.52	28.28
Onions	8	4.13	0.91	1.00	21.93	24.12
Raisins	8	7.95	0.67	0.67	8.41	8.41
Sugar beet fiber	10	20.65	0.80	1.35	3.88	6.52

Table 993.19C. Method performance for insoluble dietary fiber (IDF) in foods by enzymatic-gravimetric method (phosphate buffer)

Food	No. of labs	IDF, av. %	s_r	s_R	RSD_r, %	RSD_R, %
Apricots	8	13.89	0.73	1.08	5.24	7.80
Carrots	9	12.66	0.59	1.27	4.68	10.03
Chickpeas	10	14.72	1.28	2.19	8.70	14.86
Onions	8	12.55	0.69	0.91	5.53	7.29
Raisins	10	23.03	0.90	1.37	3.91	5.95
Sugar beet fiber	10	45.57	0.67	0.98	1.48	2.17

able filtration times, both when the IDF fraction was removed from the SDF fraction by filtration and when the precipitated SDF fraction was filtered in the second filtration step. Several laboratories suggested that these problems could be overcome by using 0.5–0.25 g test portions for analysis of materials containing high levels of viscous fiber, which hinder filtration. The high variability in the determination of SDF was also due in part to the lack of uniformity of the Celite used by the collaborators. We therefore sought to minimize this problem by sending the collaborating laboratories a specific Celite preparation of uniform quality (same lot number) and instructions for preparation before use.

All 10 laboratories submitted results. In a few cases, no results were reported because the investigator did not have sufficient test sample (apricots and onions, IDF and SDF), and in other cases some of the analytes were lost during analysis (chickpeas, SDF; raisins, 1 of a pair of duplicates of TDF, IDF, and SDF).

The results of the individual determinations for TDF are shown in Table 1. For the TDF determination, only 1 laboratory's value was a Cochran standard statistical outlier for 1 food (carrots), and 2 laboratories' values were paired Grubbs outliers for onions and sugar beet fiber. All values were used in the statistical calculation, however. All other values were used as reported by the analysts. The measures of precision for TDF are shown in Table **993.19B**. The average TDF values ranged from 14.33% for chickpeas to 66.07% for sugar beet fiber. The repeatability relative standard deviation (RSD_r) for the determination of TDF of the 6 foods analyzed ranged from 1.74% for sugar beet fiber to 5.93% for chickpeas. The reproducibility relative standard deviation (RSD_R) ranged from 2.41% for sugar beet fiber to 7.01% for chickpeas. The RSD_R values for the 6 foods averaged 4.98%, which is considered excellent for this determination and a considerable improvement over our results in the 1988 collaborative study (4).

Table 993.19B. Method performance for total dietary fiber (TDF) in foods by enzymatic-gravimetric method (phosphate buffer)

Food	No. of labs	TDF av., %	s_r	s_R	RSD_r, %	RSD_R, %
Apricots	10	24.63	0.74	0.97	3.03	3.95
Carrots	9	23.25	0.49	0.79	2.10	3.38
Chickpeas	10	14.33	0.85	1.00	5.93	7.01
Onions	9	16.13	0.95	1.02	5.88	6.33
Raisins	10	30.28	0.81	2.05	2.69	6.78
Sugar beet fiber	10	66.07	1.15	1.59	1.74	2.41

The results of the individual determinations of IDF are shown in Table 1. Because there were no statistical outliers, all values were used as reported. The measures of precision for the determination of IDF are shown in Table **993.19C**. The average IDF values for the products analyzed ranged from 12.55% for onions to 45.57% for sugar beet fiber. RSD_r values ranged from 1.48% for sugar beet fiber to 8.70% for chickpeas. RSD_R values ranged from 2.17% for sugar beet fiber to 14.86% for chickpeas, with an average RSD_R of 8.02% for the 6 foods. This compares very favorably with the average RSD_R for IDF of 11.09% in the previous collaborative study in which the IDF method was adopted official first action.

The results of the individual determinations of SDF are also shown in Table 1. Only 1 value was a Grubbs outlier and was not used in the statistical calculations. The measures of precision for the determination of SDF are shown in Table **993.19A**. The products analyzed had average SDF values that ranged from 1.21% for chickpeas to 20.65% for sugar beet fiber. RSD_r values ranged from 3.78% for apricots to 21.93% for onions. RSD_R values ranged from 6.52% for sugar beet fiber to 28.28% for chickpeas. There was, however, a substantial reduction in the RSD_R values, in all cases, when these results were compared with those from the previous collaborative study (4). The RSD_R value for apricots was decreased from 16.36 to 8.11%, for carrots from 15.76 to 9.61%, for chickpeas from 44.38 to 28.28%, for onions from 37.86 to 24.12%, and for raisins from 41.21 to 8.41%.

Table 2 shows the method performance for the ratio of TDF determined by the sum of IDF and SDF to the independent determination of TDF. The ratio varied from 1.00 for sugar beet fiber to 1.10 for chickpeas, with an average ratio of 1.03. The RSD_R values ranged from 2.27% for sugar beet fiber to 11.46% for onions, with an average RSD_R of 6.85%. This means that,

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Table 2. Method performance for ratio of total dietary fiber (TDF) determined by sum of insoluble dietary fiber (IDF) and soluble dietary fiber (SDF) to independent determination of TDF

Food	No. of labs	Ratio, (IDF + SDF) /TDF[a]	RSD_r, %	RSD_R, %
Apricots	8	1.02	4.05	5.26
Carrots	9	1.03	4.25	4.93
Chickpeas	9	1.10	5.77	11.15
Onions	9	1.02	8.49	11.46
Raisins	8	1.01	1.44	6.00
Sugar beet fiber	9	1.00	2.13	2.27

[a] Independent determination.

to determine TDF, the final action method for TDF can be used or the values determined for SDF and IDF can be summed.

Recommendation

We recommend that the enzymatic-gravimetric method for the determination of soluble dietary fiber be adopted first action. This method can be used to determine total dietary fiber in conjunction with the official first action method for insoluble dietary fiber. The final action method for total dietary fiber may be used when only the total is required.

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